Cloud Computing (COM467)

Unit 2

Cloud Computing Architecture

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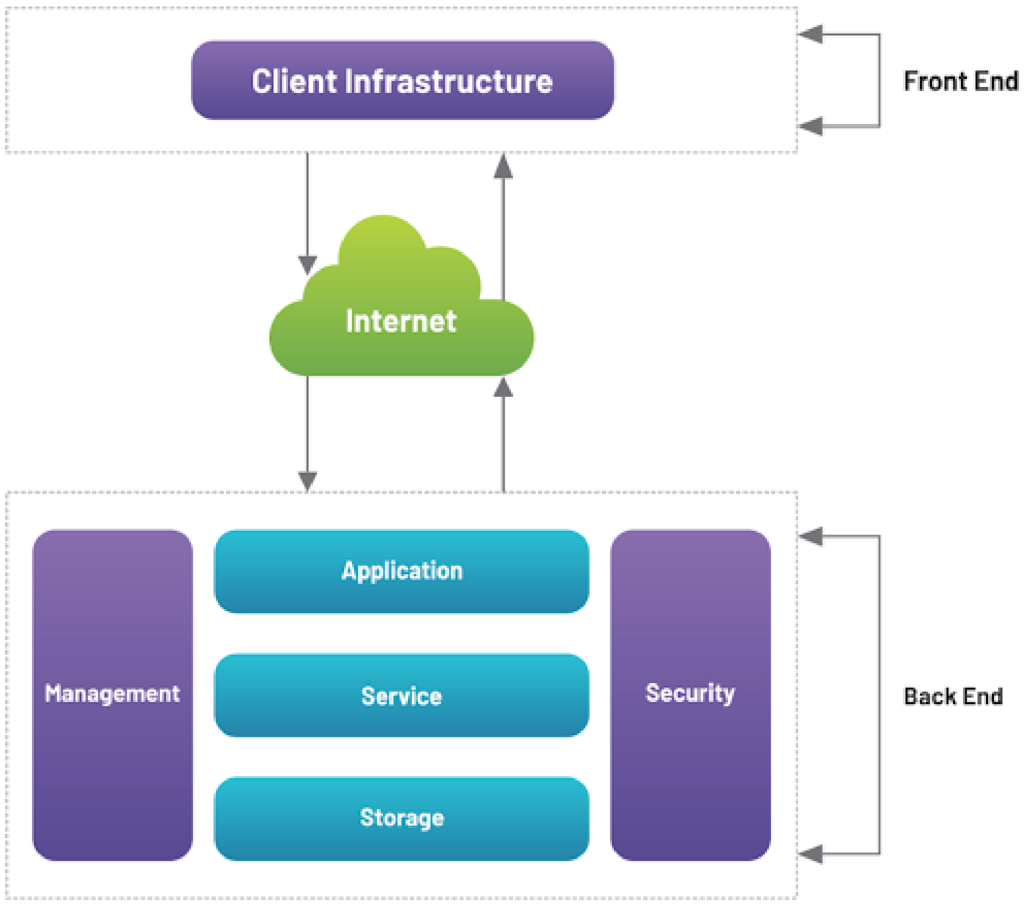
CHAPTER CONTENTS

- Platform as Service
- Software as a Service
- Infrastructure as Service
- Public Cloud
- Private Cloud
- Hybrid Cloud
- Community Cloud
- Cloud Design and Implementation using SOA,
- Security, Trust and Privacy

CLOUD COMPUTING ARCHITECTURE

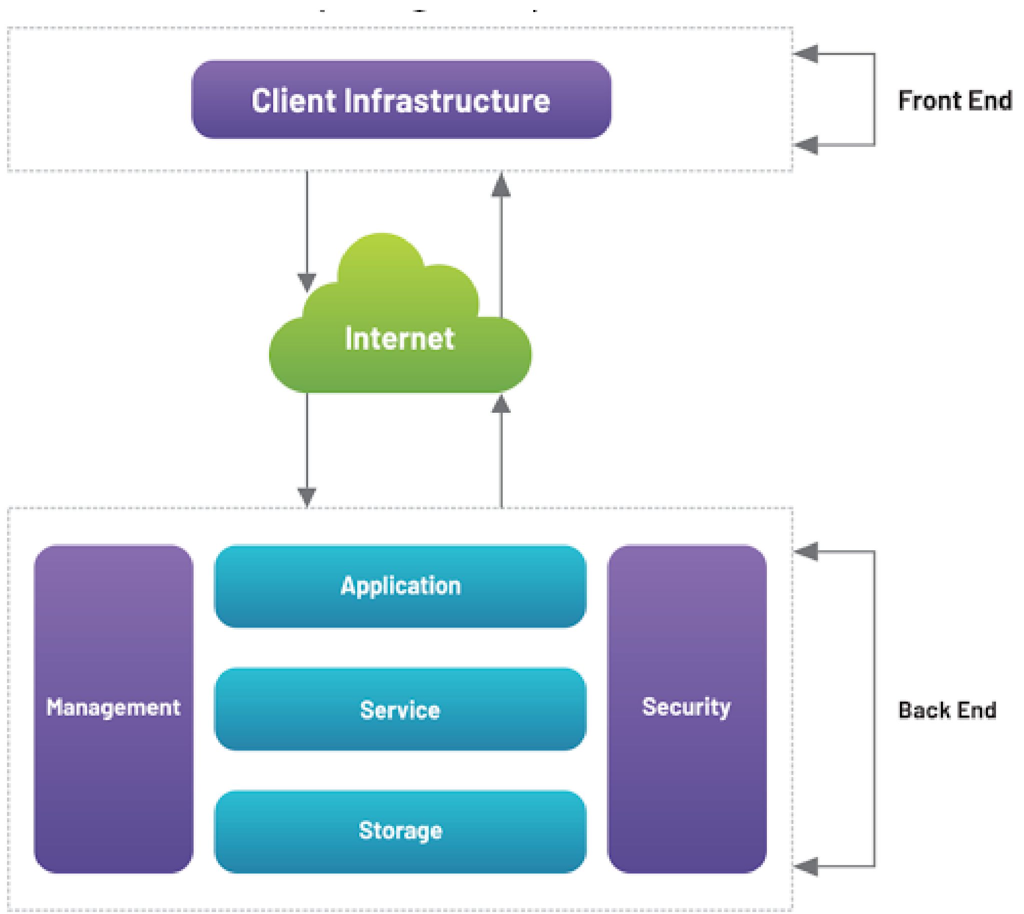
- **Cloud Computing Architecture** defines the **structure and components** of a cloud system.
- It provides a **blueprint** for designing, deploying, and managing cloud-based applications and services.
- Cloud Computing Architecture is divided into two parts :
    - front-end (client side)
    - back-end (cloud infrastructure)
- Front-end and back-end communicate via a network or internet to deliver cloud services.



CLOUD COMPUTING ARCHITECTURE

a. Front-End

- This is what the **user interacts with** — it represents the **client's interface** with the cloud.
- Provides applications and the interfaces required for the cloud- based service.
- **Components of front-end are:**
  - **User Interface (UI):** The web browser or application used to access cloud services (e.g., web portal, mobile app).
  - **Client Devices:** Computers, smartphones, tablets, etc., that connect to the cloud.
  - **Client Software:** Software or web applications used to interact with cloud services (e.g., browser for SaaS, custom API client for IaaS).



CLOUD COMPUTING ARCHITECTURE

b. Back-End

- The **Back-End** is the core of cloud computing, and it includes everything that happens behind the scenes to make cloud services function smoothly.
- It is responsible for
  - running and monitoring cloud applications
  - Storing and processing huge amounts of data
  - Managing cloud infrastructure (servers, storage, networking)
  - Keeping data and systems secure
- The components of the back-end cloud architecture are mentioned below.
  1. **Application-** Software or a platform which provides the result to the end- user (with resources).
  2. **Service** – Provide utility in the architecture, Widely used services are storage application development environments and web services
  3. **Storage -** It stores and maintains data like files, videos, documents, etc. over the internet.
     Eg: Amazon S3, Oracle Cloud-Storage, Microsoft Azure Storage
  4. **Management -** Its task is to allot **specific resources** to a **specific task** and **establishes coordination** among the cloud resources. It helps in the management of components like application, task, service, security, data storage, and cloud infrastructure.
  5. **Security -** Implements security management to the cloud server with virtual firewalls . It provides secure cloud resources, systems, files, and infrastructure to end-users

CLOUD ARCHITECTURE COMPONENTS

- Components of Cloud Computing architecture :
  - **Hypervisor**
  - **Management Software**
  - **Deployment Software**
  - **Network**
  - **Cloud Server**
  - **Cloud Storage**

CLOUD COMPUTING ARCHITECTURE COMPONENTS

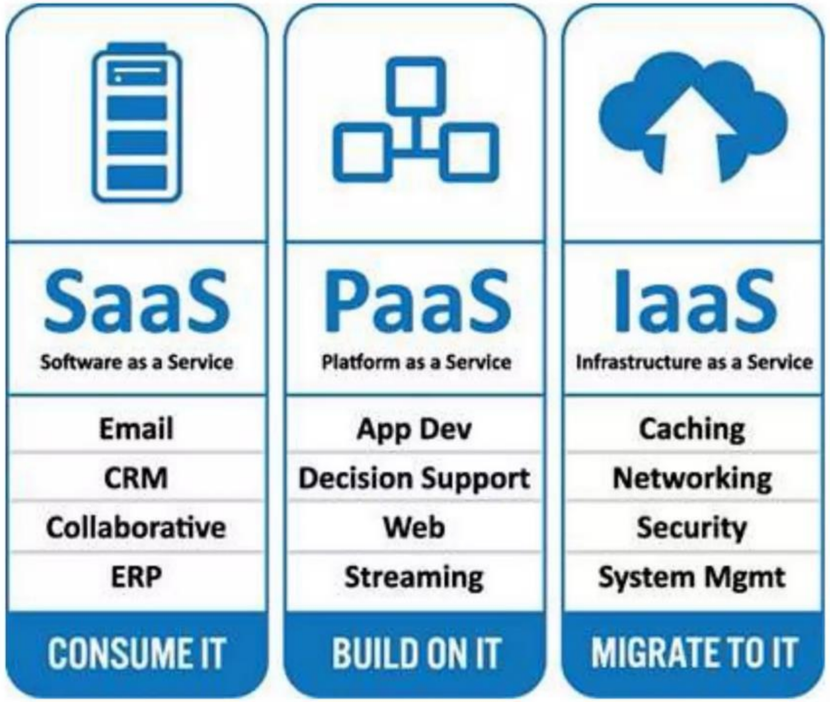
- **Hypervisor**
  - It is a **virtual machine monitor** which provides **Virtual Operating Platforms** to every user
  - Manages guest operating systems in the cloud
  - It runs a separate virtual machine on the back end which consists of software and hardware
  - Its main objective is to **divide** and **allocate** resources
- **Management Software**
  - Its responsibility is to manage and monitor cloud operations with various strategies to increase the performance of the cloud
  - Some of the operations performed by the management software are: **compliance**, **auditing management of overseeing disaster contingency plans**
- **Deployment Software**
  - It consists of all the mandatory installations and configurations required to run a cloud service
  - All deployment of cloud services is performed using a **deployment software**
  - The three different models which can be deployed are: SaaS, PaaS, IaaS

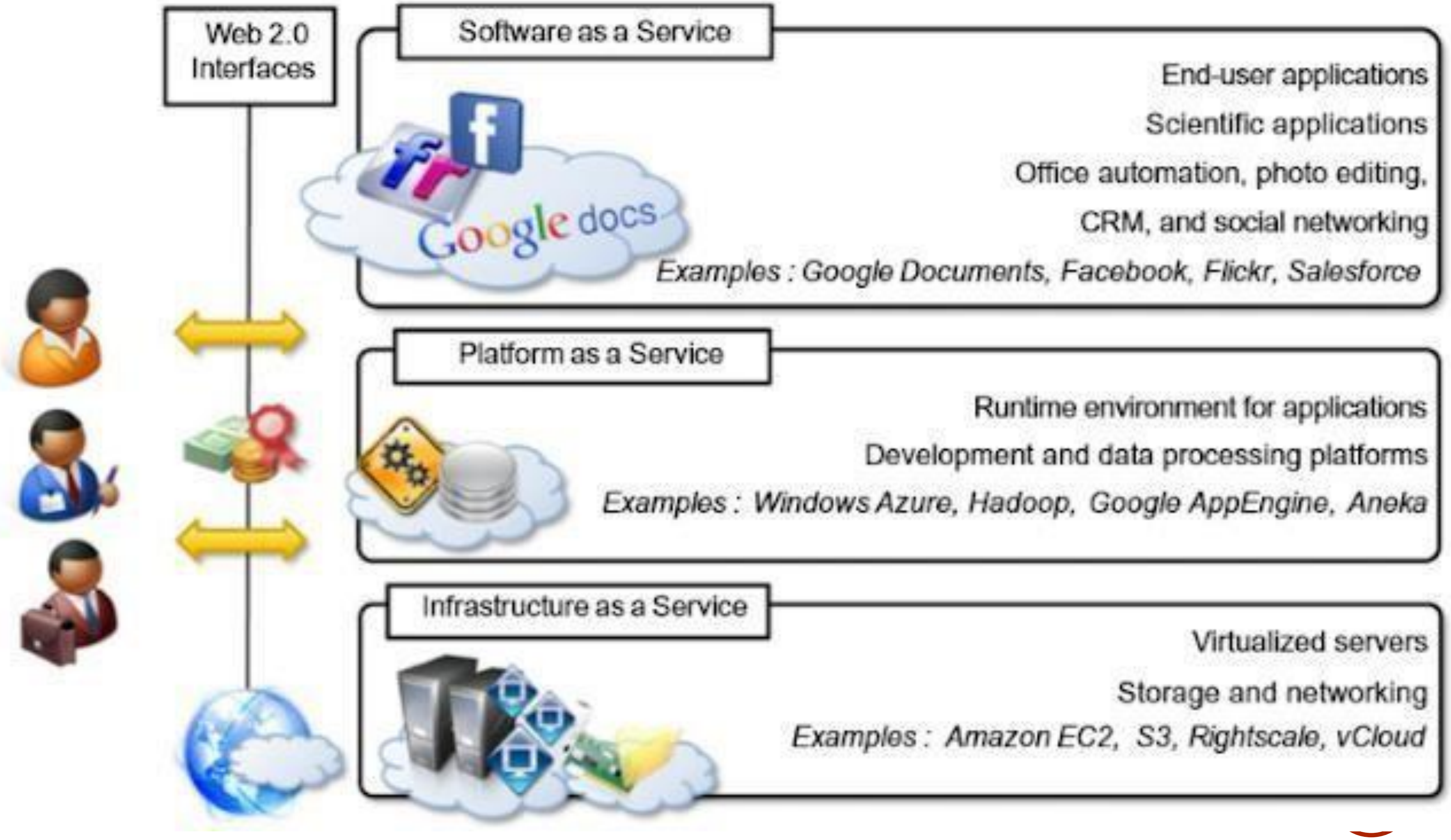
CLOUD ARCHITECTURE COMPONENTS

- **Network**
  - It connects the front-end and back-end.
  - Allows every user to access cloud resources
  - Helps users to connect and customize the route and protocol
- **Cloud Server**
  - It is a virtual server which is hosted on the cloud computing platform
  - It is highly flexible, secure, and cost-effective
- **Cloud Storage**
  - Every bit of data is stored and accessed by a user from anywhere over the internet
  - It is scalable at run-time and is automatically accessed
  - Data can be modified and retrieved from cloud storage over the web

CLOUD REFERENCE MODELS

- The reference model for cloud computing is an abstract model that characterizes and standardizes a cloud computing environment by partitioning it into abstraction layers and cross- layer functions.

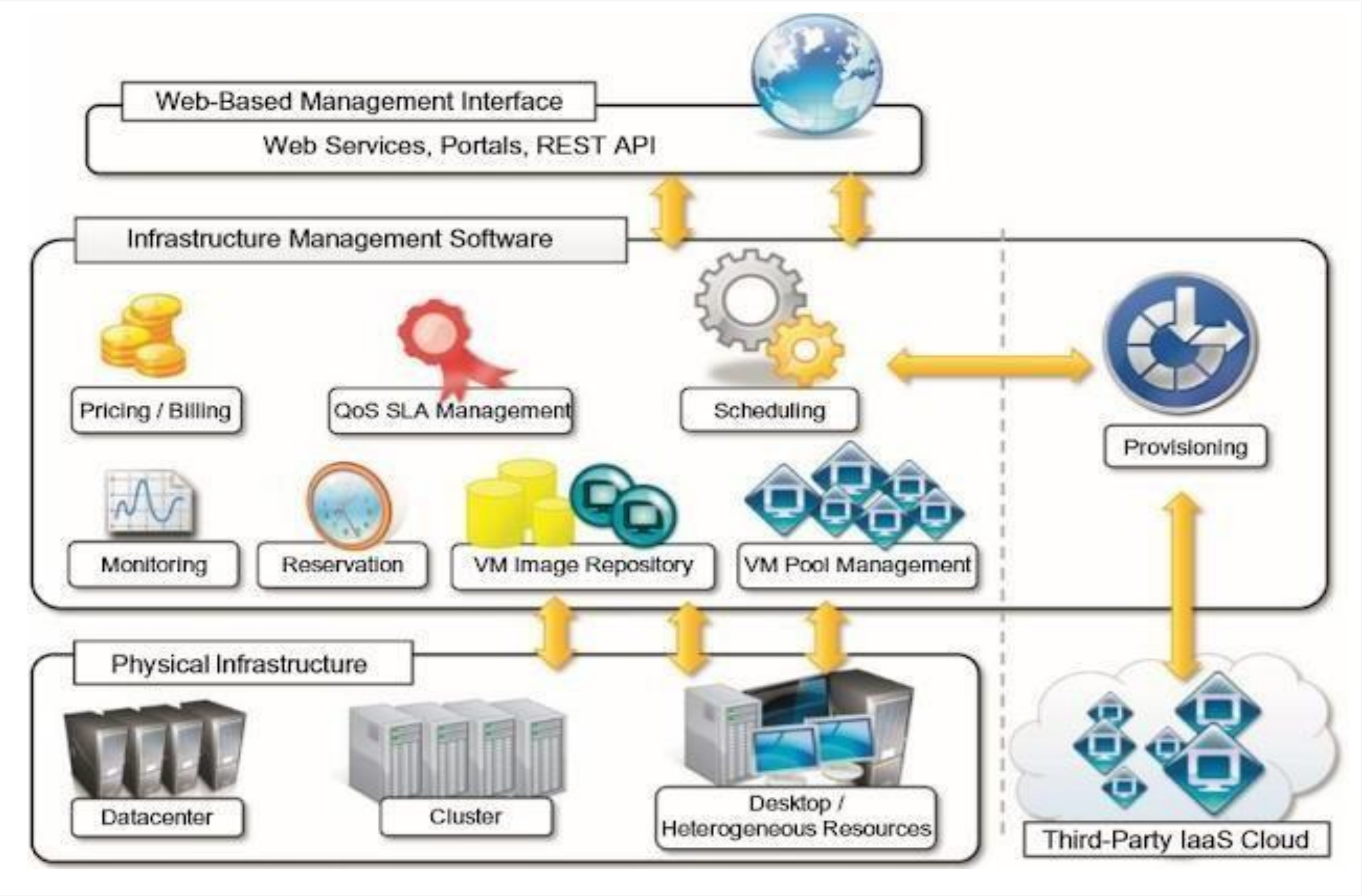




INFRASTRUCTURE AS A SERVICE (IAAS)

- IaaS provides **virtualized computing resources** over the internet.
- It gives users access to **servers, storage, networking, and virtualization**, but the user has to manage the OS, applications, and data.
- IaaS examples can be categorized in two categories
  a. IaaS Physical infrastructure
  b. IaaS Management layer
- The IaaS Physical infrastructure includes the actual hardware resources in data centers such as physical servers, storage devices, networking equipment, cooling systems, and power supply.
  - These components provide the raw computing power that cloud providers virtualize to offer services to customers.
- The **IaaS Management Layer** is the software layer that controls and manages this hardware.
  - It includes virtualization platforms (like VMware, Xen, KVM), cloud management tools (OpenStack, CloudStack), orchestration systems, APIs, dashboards, monitoring, and billing tools.
  - This layer enables users to create virtual machines, allocate resources, manage networks, and interact with cloud services easily.

INFRASTRUCTURE AS A SERVICE (IAAS)



INFRASTRUCTURE AS A SERVICE (IAAS)

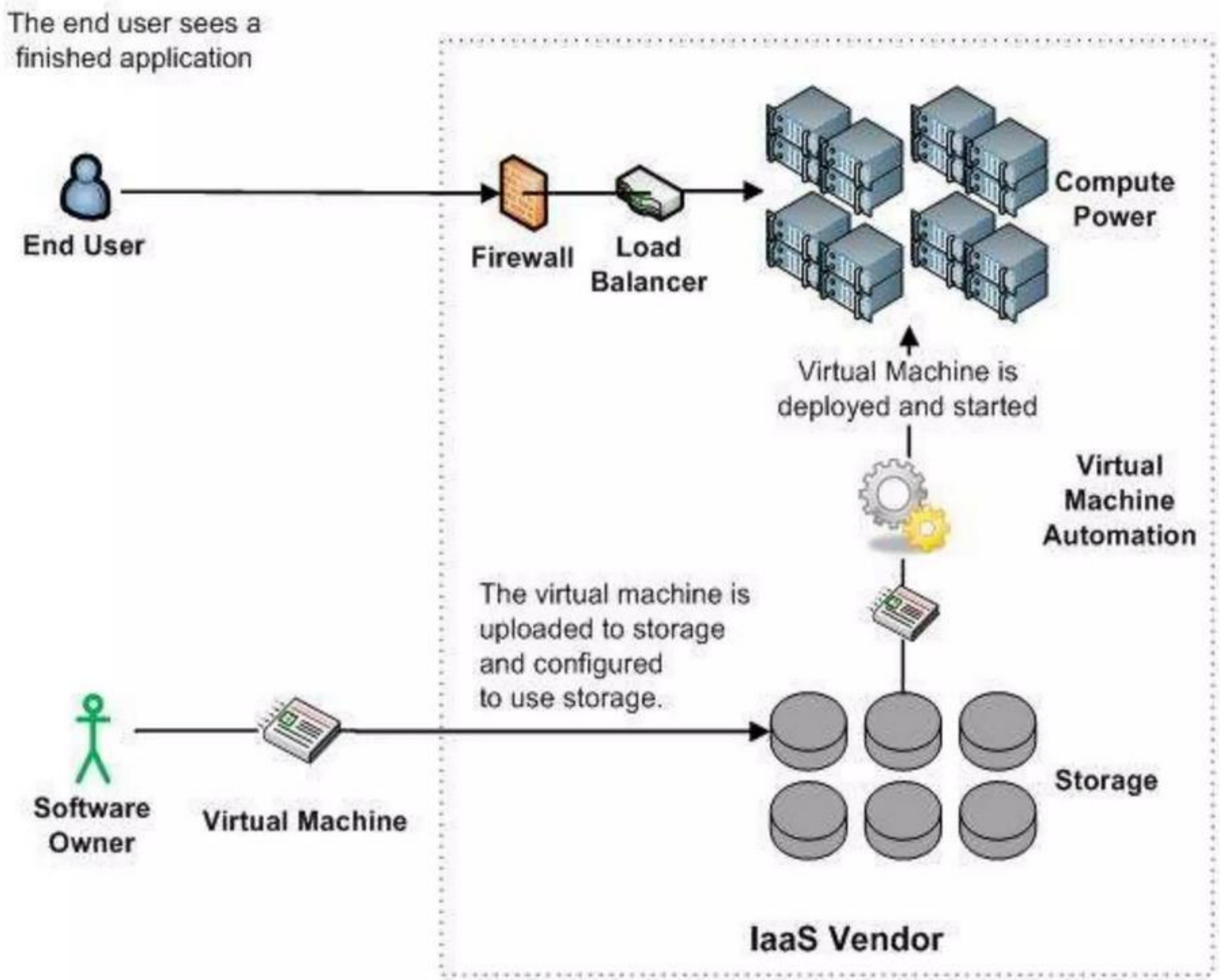
- **What the provider manages**
  - Hardware (servers)
  - Storage
  - Networking
  - Virtualization layer
- **What the user manages**
  - Operating System
  - Middleware
  - Applications
  - Data
  - Security configurations
- **Examples**
  - Amazon EC2
  - Microsoft Azure Virtual Machines
  - Google Compute Engine
- **Use Case**

  Useful for system administrators who want **full control** over the computing environment.

INFRASTRUCTURE AS A SERVICE (IAAS)

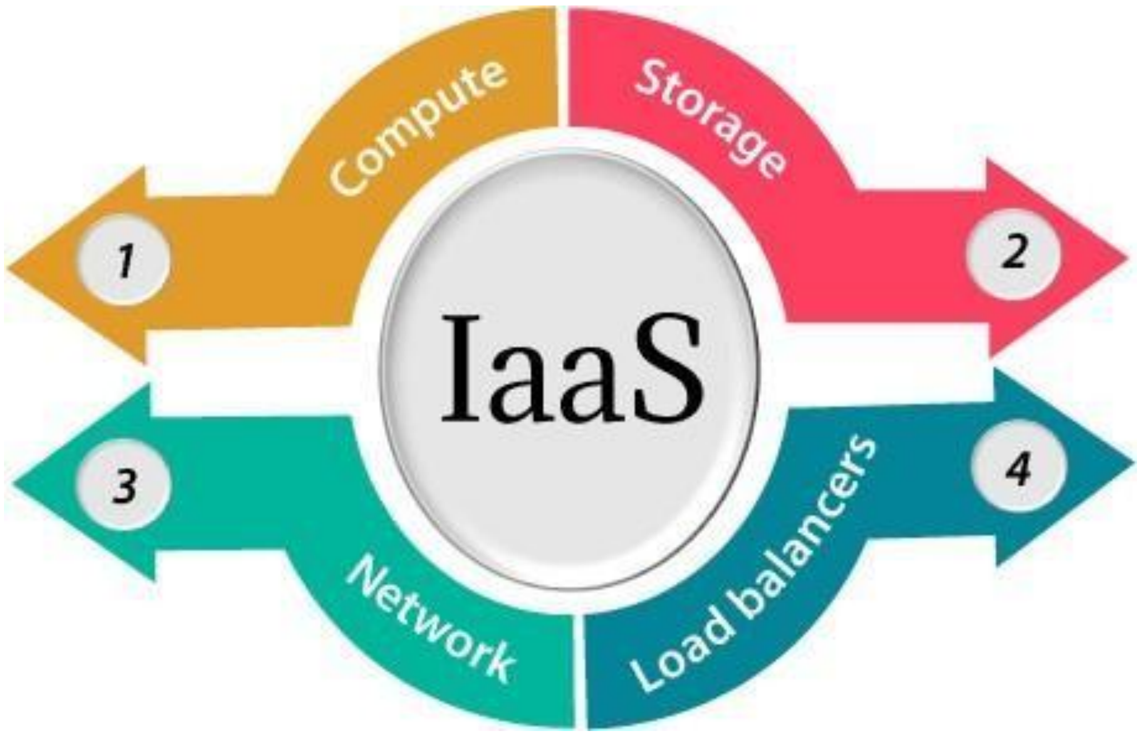
- Characteristics of IaaS
  - Resources are available as a service
  - Services are highly scalable
  - Dynamic and flexible
  - GUI and API-based access
  - Automated administrative tasks

Infrastructure as a Service (IaaS)



INFRASTRUCTURE AS A SERVICE (IAAS)

- IaaS provider provides the following services -
  1. **Compute:**
     Computing as a Service includes virtual central processing units and virtual main memory for the VMs that is provisioned to the end- users.
  2. **Storage:**
     IaaS provider provides back-end storage for storing files.
  3. **Network:**
     Network as a Service (NaaS) provides networking components such as routers, switches, and bridges for the VMs.
  4. **Load balancers:**
     A load balancer evenly distributes incoming traffic across multiple servers to prevent overload and improve performance.
     It also detects unhealthy servers and redirects traffic to healthy ones to ensure high availability.



INFRASTRUCTURE AS A SERVICE (IAAS)

Advantages of IaaS Cloud Computing layer

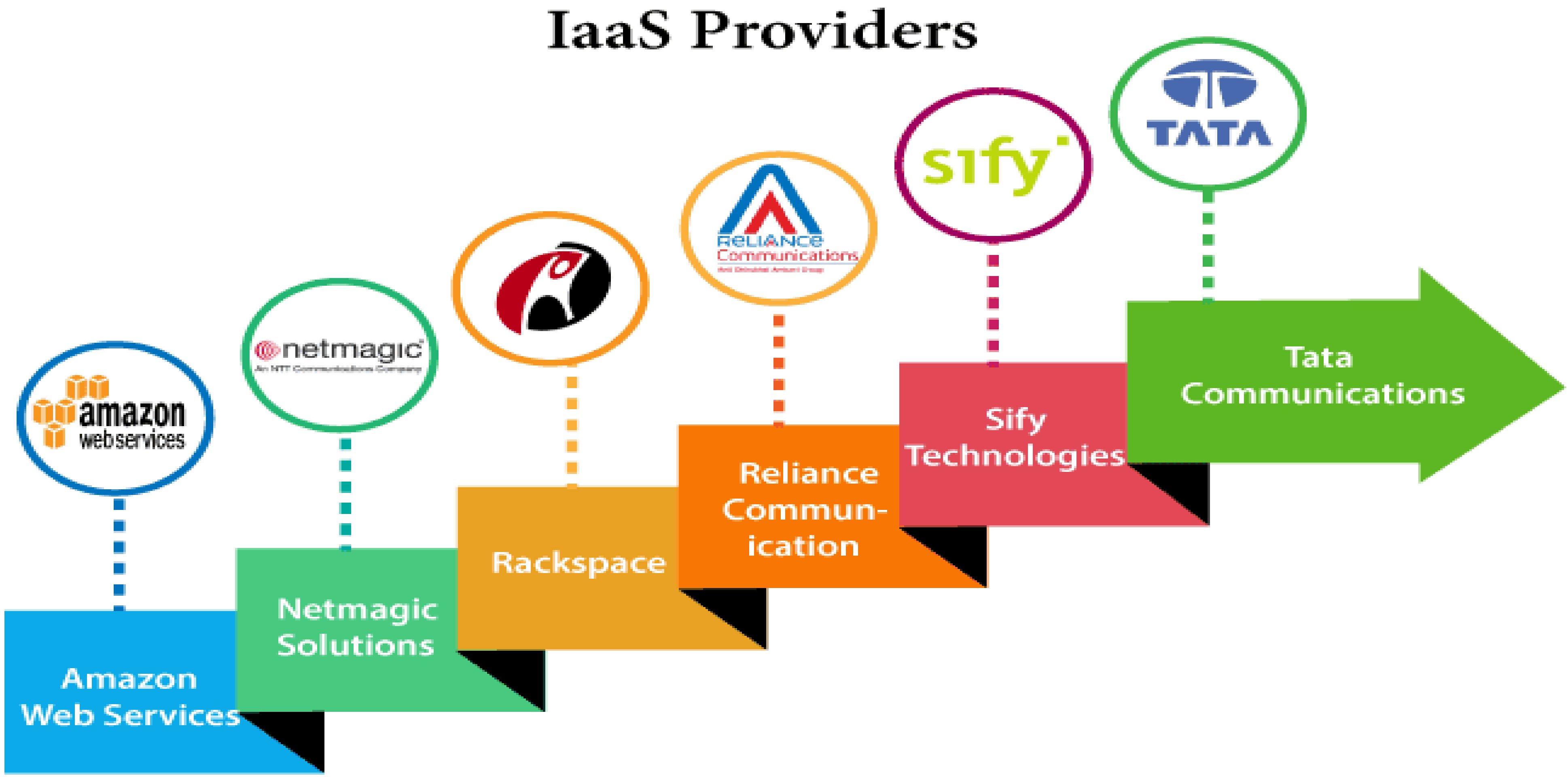
1. **Shared Infrastructure** - Allows multiple users to share the same physical infrastructure.
2. **Web access to the Resources** - Allows IT users to access resources over the internet.
3. **Pay-as-per-use Model** - IaaS providers provide services based on the pay-as-per-use basis.
4. **Focus on the Core Business** - IaaS providers focus on the organization's core business rather than on IT infrastructure.
5. **On-demand Scalability -** Users do not worry about to upgrade software and troubleshoot the issues related to hardware components.

- **Disadvantages of IaaS Cloud Computing layer**

1. **Security -** Most of the IaaS providers are not able to provide 100% security.
2. **Maintenance & Upgrade -** Although IaaS service providers maintain the software, but they do not upgrade the software for some organizations.
3. **Interoperability issues -** It is difficult to migrate VM from one IaaS provider to the other, so the customers might face problem related to vendor lock-in.

INFRASTRUCTURE AS A SERVICE (IAAS)

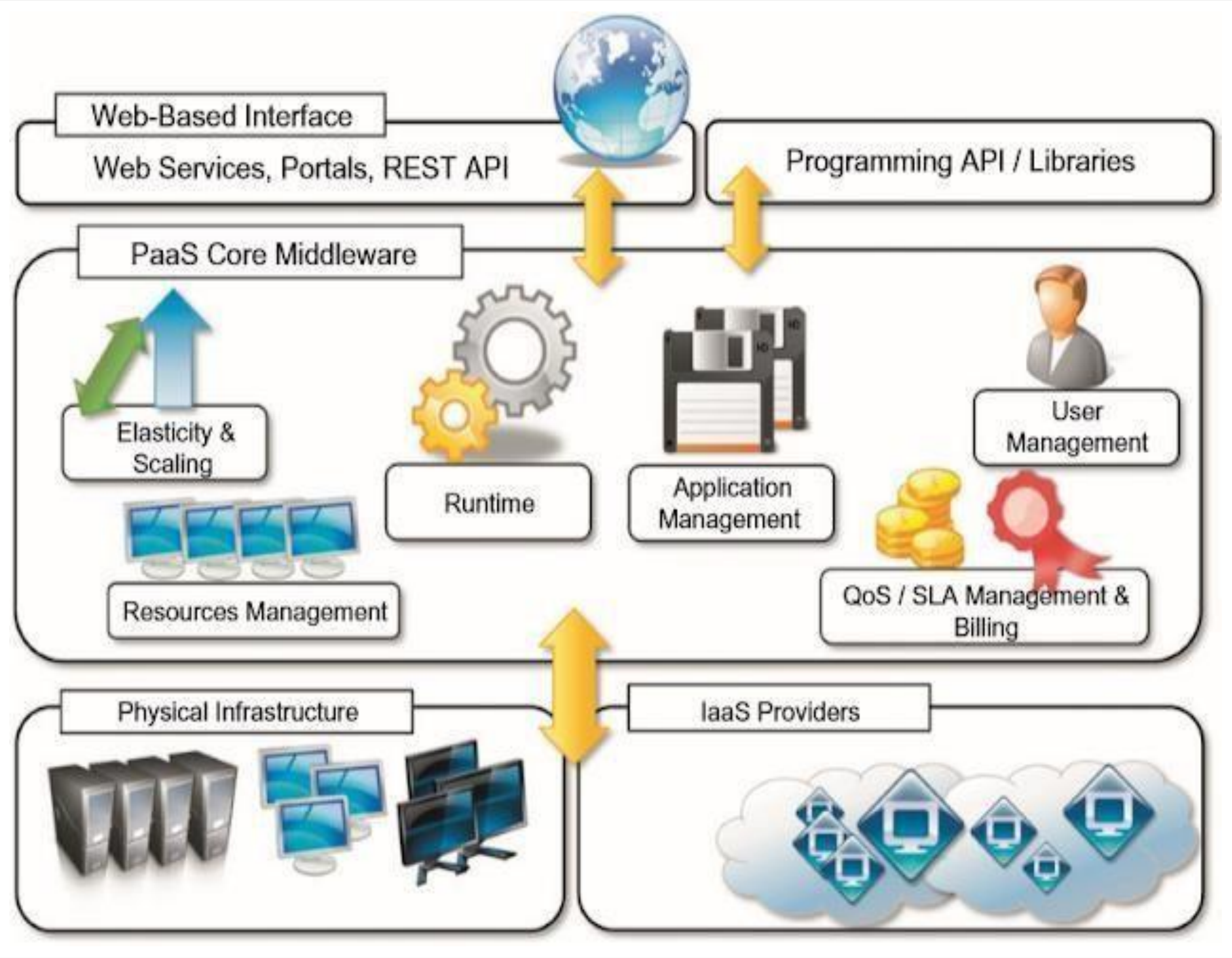
Top IaaS Providers who are providing IaaS Cloud Computing platform



PLATFORM AS A SERVICE (PAAS)

- PaaS provides a **platform** for developers to build, run, and deploy applications **without managing the underlying infrastructure**.
- The provider manages servers, OS, and runtime environments so developers can focus only on coding.
- It constitute the **middleware** on top of which applications are built.
- **Application management** is the core functionality of the middleware.
- Provides run time environments for the applications.
- PaaS provides
  - **a. Applications deployment**
  - **b. Configuring application components**
  - **c. Provisioning and configuring supporting technologies**
- For users PaaS interfaces can be in the form of a Web-based interface or in the form of programming APIs and libraries.
- PaaS solutions generally include the infrastructure as well.
- Pure PaaS offered only the user-level middleware.
- Some examples: Google App Engine, Azure, AWS, Force.com

PLATFORM AS A SERVICE (PAAS)



PLATFORM AS A SERVICE (PAAS)

- **PaaS classification:**

  a. PaaS-I: Runtime environment with **Web- hosted application development platform**. Rapid application prototyping.

  - For example Force.com which is a combination of middleware and infrastructure product type.

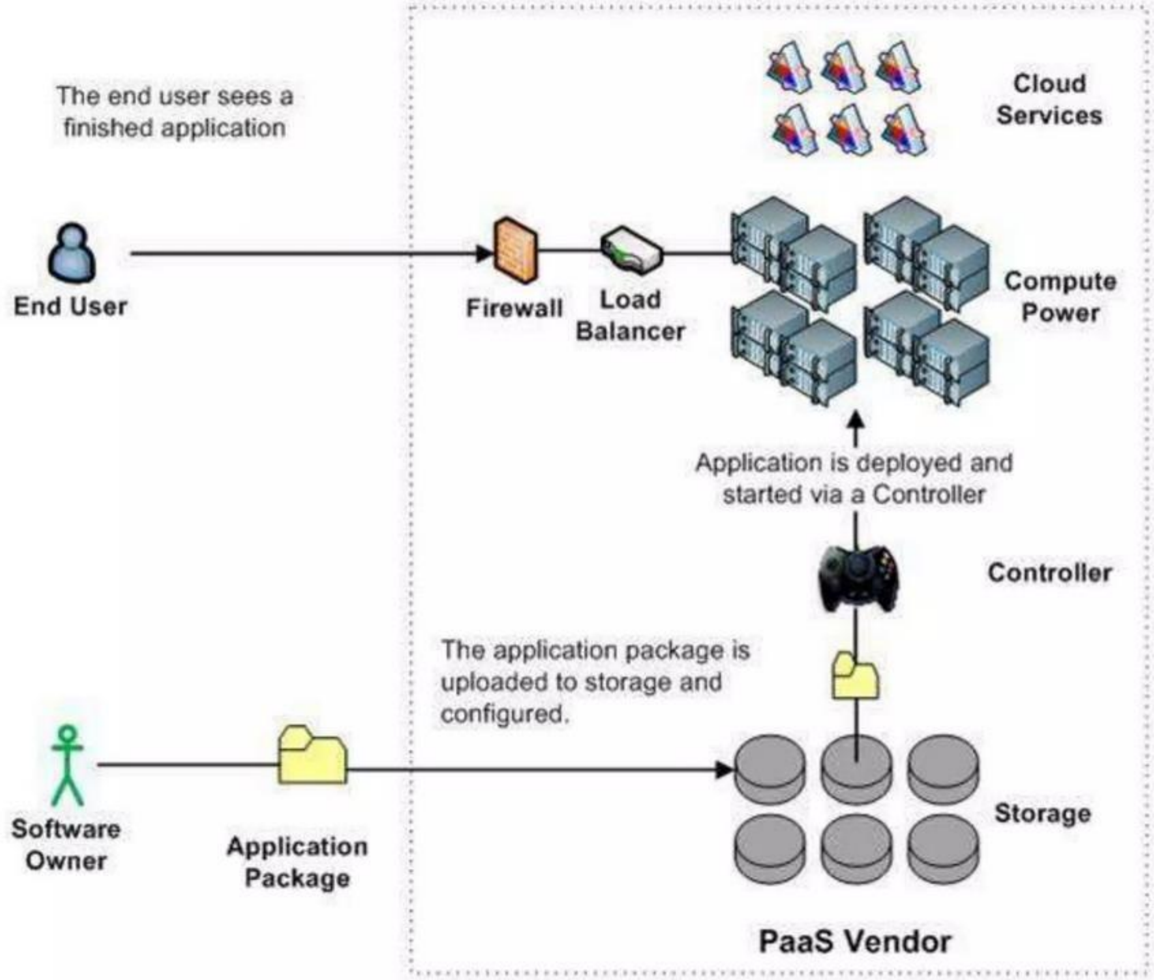
  a. PaaS-II: Runtime environment for **scaling Web applications**. The runtime could be enhanced by additional components that provide scaling capabilities.

  - For example Google AppEngine which is a combination of middleware and infrastructure product type. Appscale is middlware product type.

  c. PaaS-III: Middleware and programming model for developing **distributed applications** in the cloud.

  - For example Microsoft Azure which is a combination of middleware and infrastructure product type. Manjrasoft Aneka is a middleware product type.

Platform as a Service (PaaS)



salesforce.com Success On Demand.

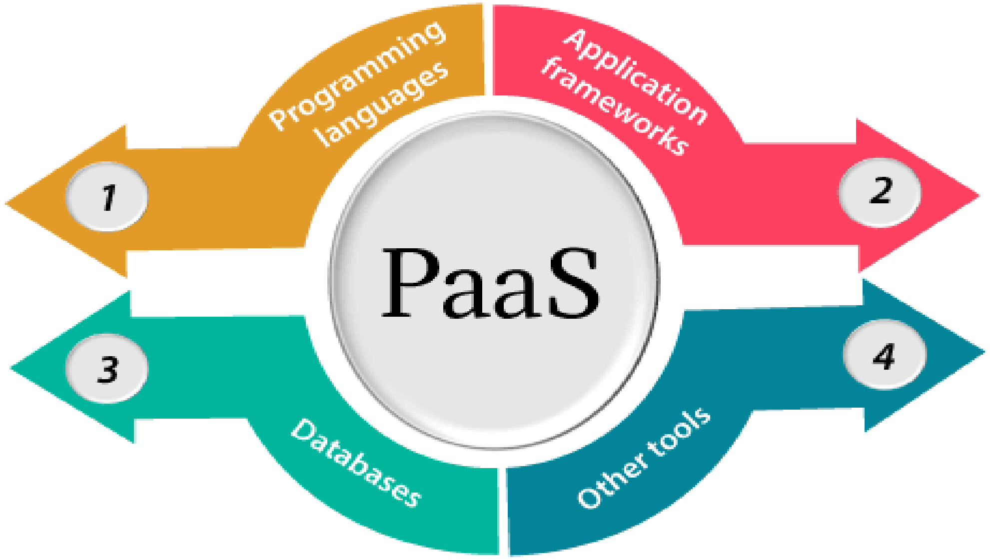
Google App Engine

Windows Azure

PLATFORM AS A SERVICE (PAAS)

- PaaS providers provide the Programming languages, Application frameworks, Databases, and Other tools:

1. **Programming languages -** PaaS providers provide various programming languages for the developers to develop the applications.
   - Eg. Java, PHP, Ruby, Perl, and Go.
2. **Application frameworks -** PaaS providers provide application frameworks to easily understand the application development.
   - Eg. Node.js, Drupal, Joomla, WordPress, Spring, Play, Rack, and Zend.
3. **Databases -** PaaS providers provide various databases such as ClearDB, PostgreSQL, MongoDB, and Redis to communicate with the applications.
4. **Other tools -** PaaS providers provide various other tools that are required to develop, test, and deploy the applications.



PLATFORM AS A SERVICE (PAAS)

- **Characteristics of PaaS:**

1. **Runtime framework:** The runtime framework executes end-user code according to the policies set by the user and the provider.
2. **Abstraction:** PaaS offer a way to deploy and manage applications on the cloud rather than a virtual machine on top of which the IT infrastructure is built and configured.
3. **Automation:** PaaS deploy the applications automatically.
4. **Cloud services:** Provide services for creation, delivery, monitoring management, reporting of applications.

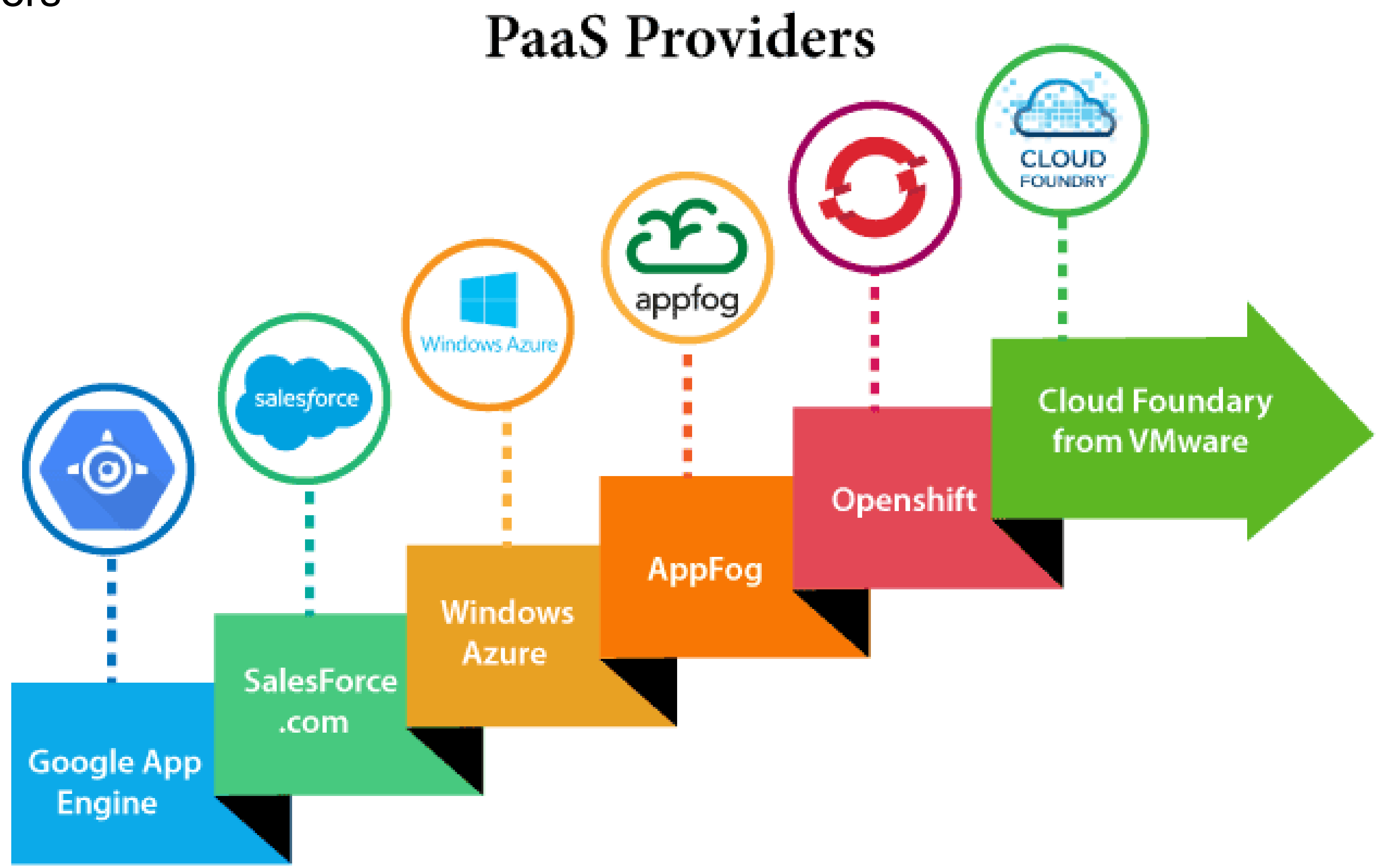
PLATFORM AS A SERVICE (PAAS)

- **Advantages of PaaS Cloud Computing layer**
  1. **Simplified Development** - Allows developers to focus on development and innovation without worrying about infrastructure management.
  2. **Lower risk** - No need for up-front investment in hardware and software.
  3. **Prebuilt business functionality** - Some PaaS vendors also provide already defined business functionality so that users can avoid building everything from very scratch and hence can directly start the projects only.
  4. **Instant community** - Provide online communities where the developer can get the ideas to share experiences and seek advice from others.
  5. **Scalability** - Applications deployed can scale without any changes to the applications.

- **Disadvantages of PaaS Cloud Computing layer**
  1. **Vendor lock-in** - The migration of an application to another PaaS vendor can become a problem.
  2. **Data Privacy** - Corporate data can be a risk in terms of privacy of data.
  3. **Integration with the rest of the systems applications** - There will be chances of increased complexity when we want to use data which in the cloud with the locally stored data.

PLATFORM AS A SERVICE (PAAS)

Popular PaaS Providers



SOFTWARE AS A SERVICE (SAAS)

- SaaS delivers **ready-to-use software applications** over the internet.
  No installation required; users simply access the software via a browser or app.
- **What the provider manages Everything**, including:
  - Infrastructure
  - Platforms
  - Applications
  - Updates, security, backups
- **What the user manages**
  - Only the **use of the application**
  - Sometimes configuration settings

SOFTWARE AS A SERVICE (SAAS)

- Allows users to connect to and use cloud-based apps over the Internet.
- SaaS is the service with which end users interact directly.
- It provides a means to free users from **complex hardware** and **software management**.
- Do not need to purchase the software and required the license.
- They simply access the application website or use application with their credentials and billing details
- Customer can customize their software.
- Application is available to the customer on demand.
- It can be considered as a “one-to-many” software delivery model.
- Applications are built as per the user needs.
- Some examples:
    a. G Suite, Office 365, Dropbox, WhatsApp

SOFTWARE AS A SERVICE (SAAS)

Services Provided by SaaS providers -

- **Business Services** -. The SaaS business services include ERP (Enterprise Resource Planning), CRM (Customer Relationship Management), billing, and sales.
- **Document Management** - SaaS document management is a software application offered by a third party (SaaS providers) to create, manage, and track electronic documents.
  - Example: Slack, Samepage, Box, and Zoho Forms.
- **Social Networks** - Social networking service providers use SaaS for their convenience and handle the general public's information.
- **Mail Services** - To handle the unpredictable number of users and load on e-mail services, many e-mail providers offering their services using SaaS.

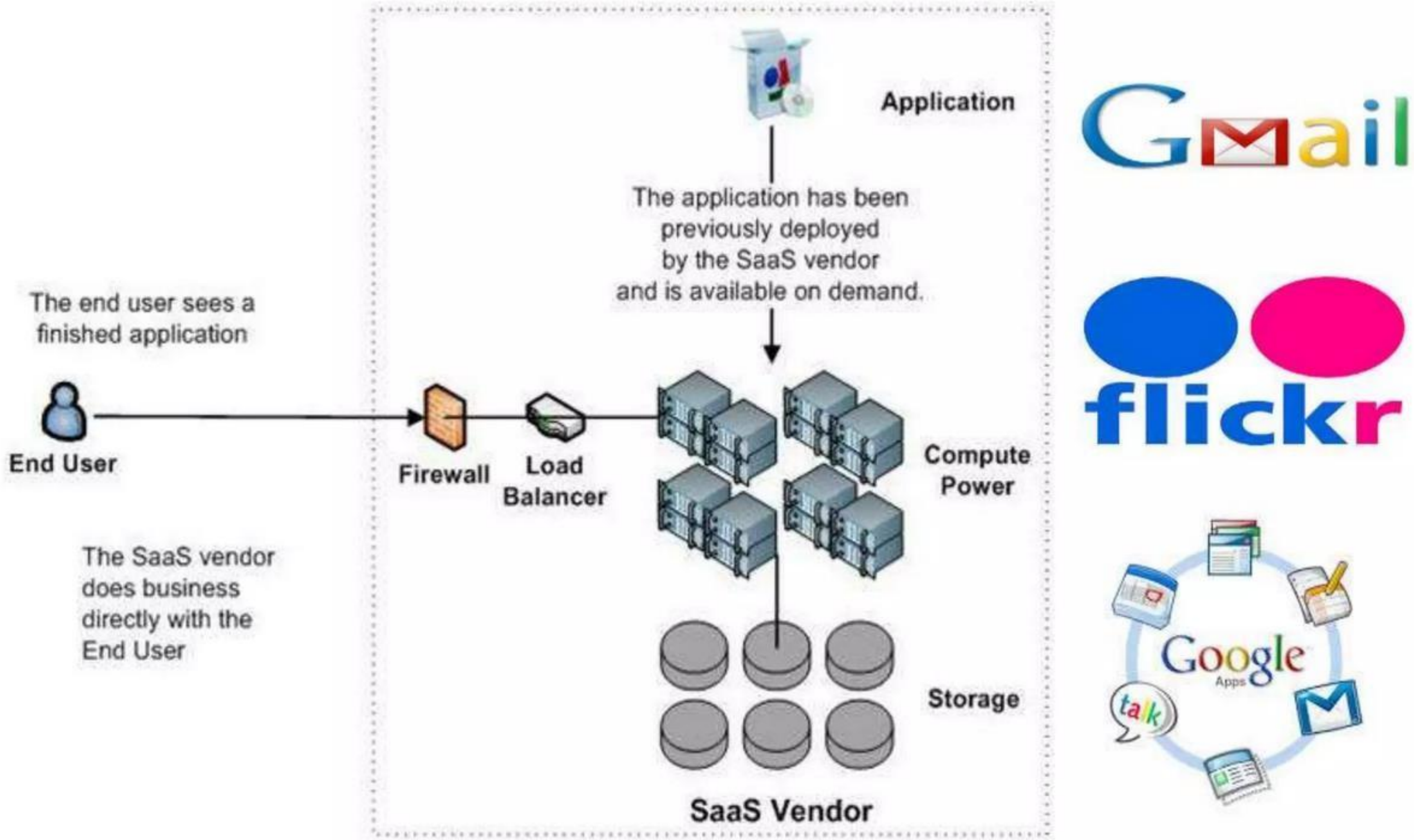
SOFTWARE AS A SERVICE (SAAS)

Advantages of SaaS Cloud Computing layer

1. **SaaS is easy to buy** - SaaS pricing is based on a monthly fee or annual fee subscription,
2. **One to Many** - SaaS services are offered as a one-to-many model means a single instance of the application is shared by multiple users.
3. **Less hardware required for SaaS** - Organizations do not need to invest in additional hardware.
4. **Low maintenance required for SaaS** - Software as a service removes the need for installation, set-up, and daily maintenance for the organizations.
5. **No special software or hardware versions required** - All users will have the same version of the software and typically access it through the web browser and reduces IT support costs by outsourcing hardware and software maintenance and support to the IaaS provider.
6. **Multidevice support** - Can be accessed from any device such as desktops, laptops, tablets, phones.
7. **API Integration** - SaaS services easily integrate with other software or services through standard APIs.
8. **No client-side installation** - SaaS services are accessed directly from the service provider using the internet connection, so do not need to require any software installation.

SOFTWARE AS A SERVICE (SAAS)

Software as a Service (SaaS)



SOFTWARE AS A SERVICE (SAAS)

Disadvantages of SaaS Cloud Computing layer

1. **Security** – Since data is stored in the cloud it is not as secure as in-house deployment.
2. **Latency issue** - There may be greater latency when interacting with the application compared to local deployment.
3. **Total Dependency on Internet** - Without an internet connection, most SaaS applications

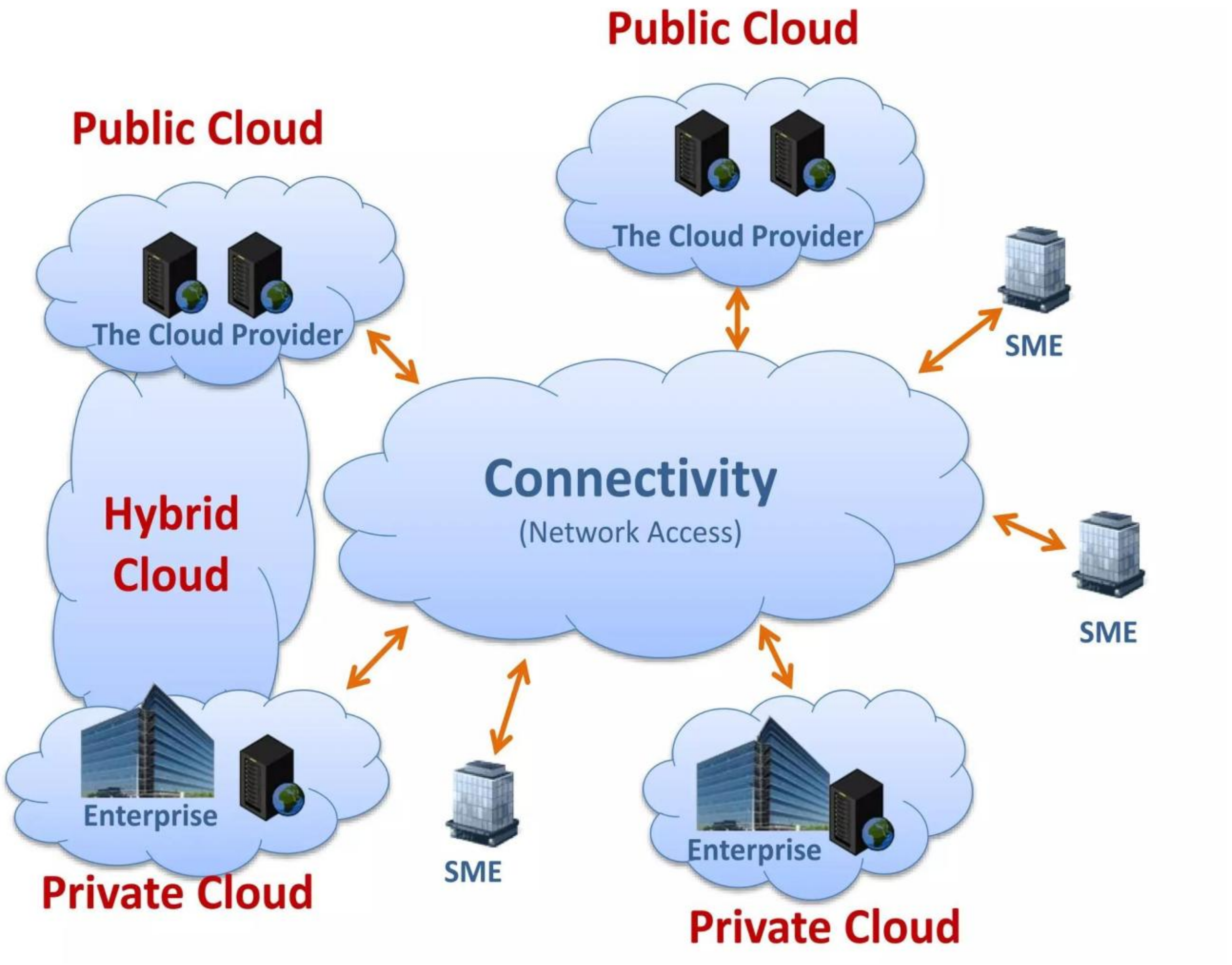
   are not usable.
4. **Switching between SaaS vendors is difficult** - Switching SaaS vendors involves the difficult and slow task of transferring the very large data files over the internet and then converting and importing them into another SaaS .

SOFTWARE AS A SERVICE (SAAS)

Popular SaaS Providers

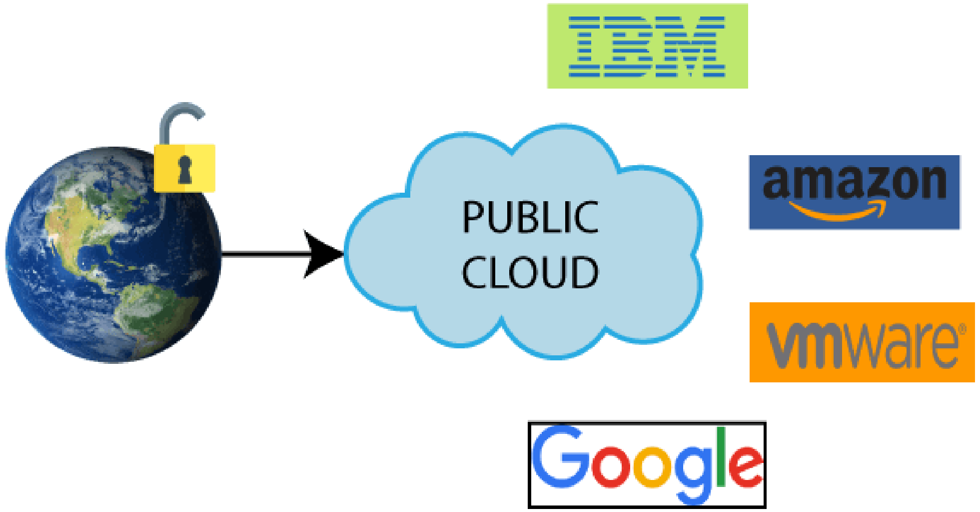
DEPLOYMENT MODELS

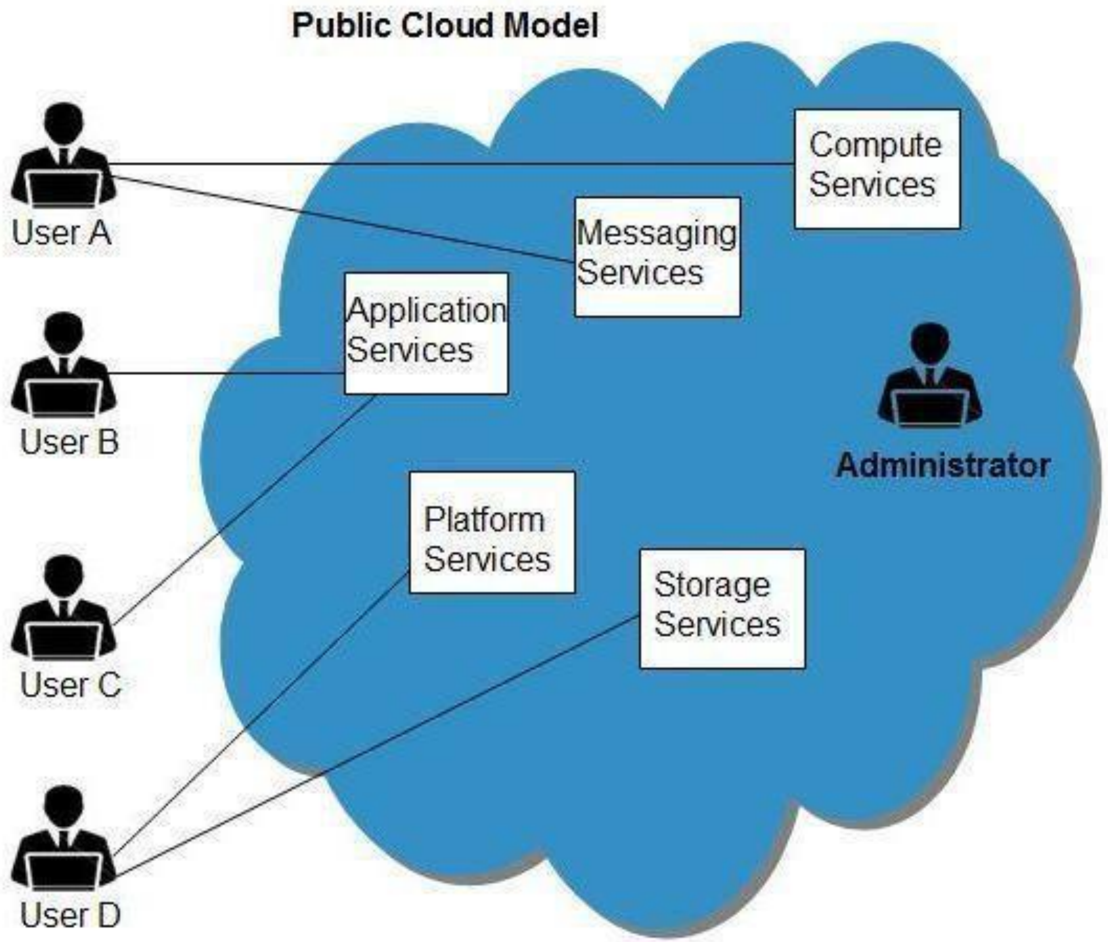
- Deployment models define the type of access to the cloud, i.e., how the cloud is located?
- Cloud can have any of the four types of access:
  - **Public**
  - **Private**
  - **Hybrid**
  - **Community**



PUBLIC CLOUD

- The public cloud allows systems and services to be easily accessible to general public.
- Eg: Amazon Elastic Compute Cloud (EC2), Microsoft Azure, IBM's Blue Cloud, Sun Cloud, and Google Cloud

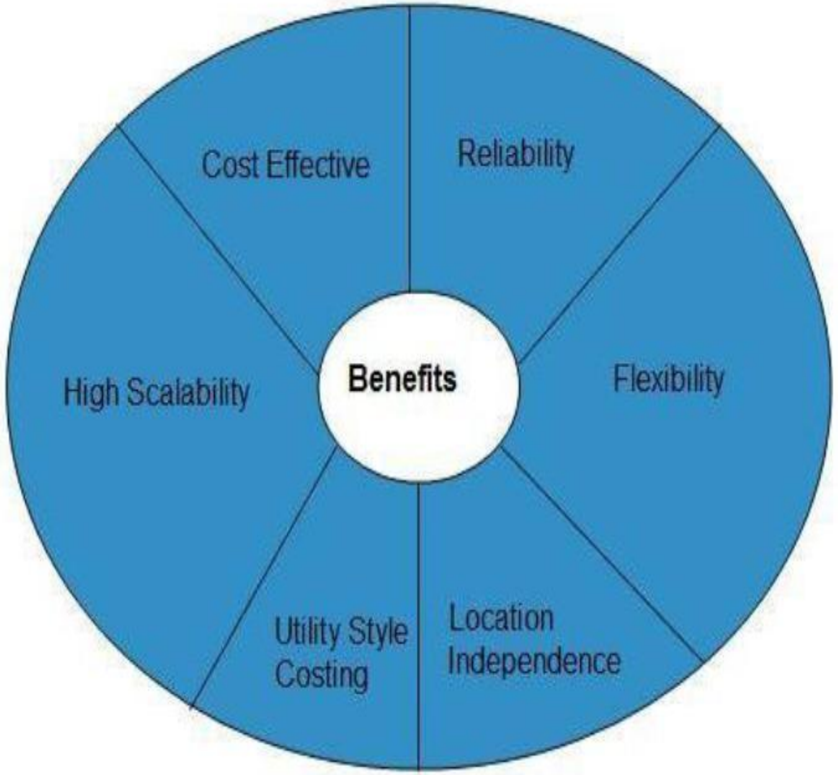




PUBLIC CLOUD

Advantages

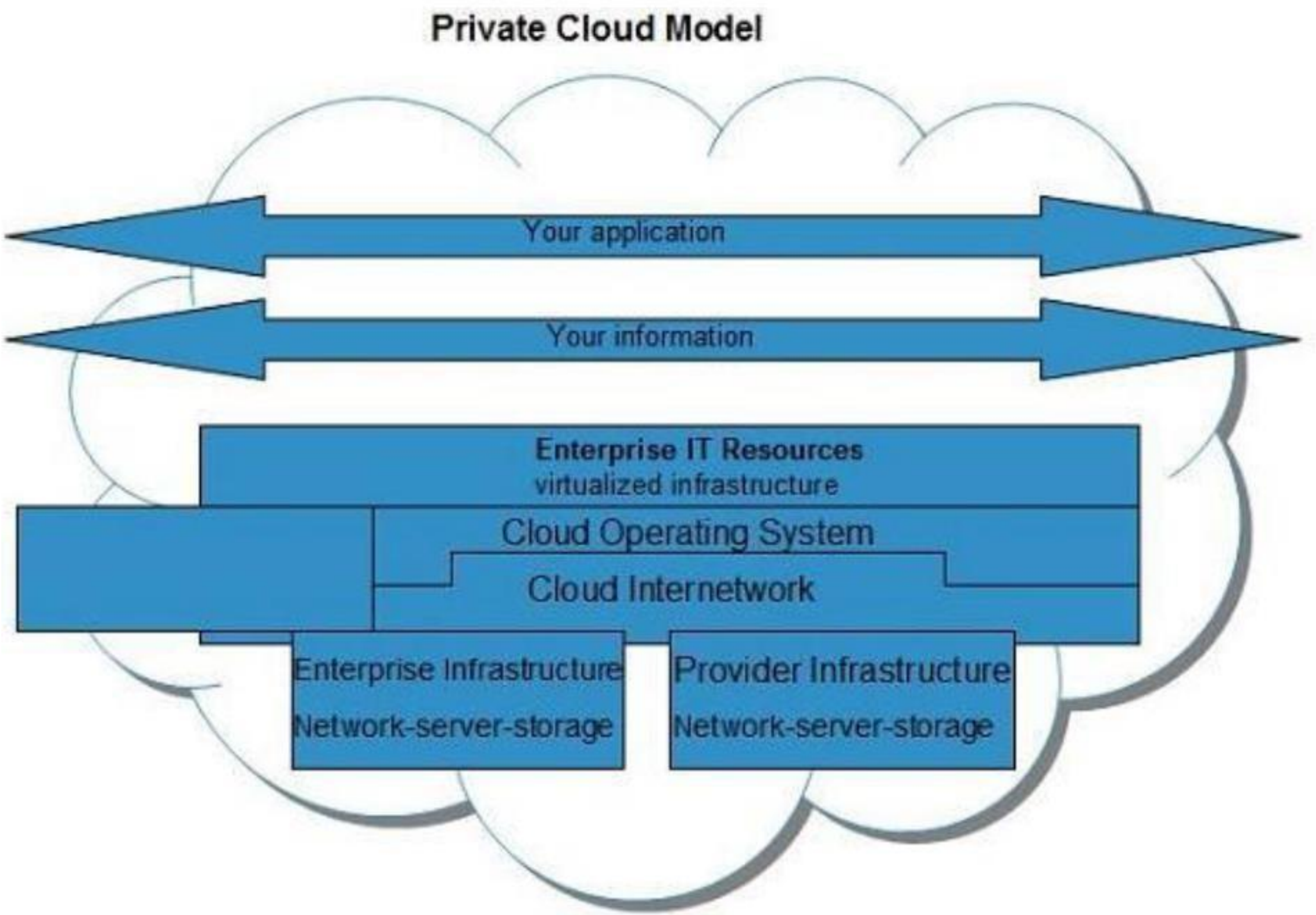
- **Cost effective -** Resources are shared with large number of consumer.
- **Reliability -** Employs large number of resources from different locations, if any of the resource fail, public cloud can employ another one.
- **Flexibility -** Easy to integrate public cloud with private cloud, hence gives consumers a flexible approach.
- **Location independence -** Services are delivered through internet, therefore ensures location independence.
- **Utility style costing -** Based on pay-per-use model .
- **High scalability -** Resources are made available on demand from a pool of resources



- **Disadvantages**
  - **Low security** - Data is hosted off-site and resources are shared publicly, therefore does not ensure higher level of security.
  - **Less customizable** - It is comparatively less customizable than private cloud.

PRIVATE CLOUD

- The private cloud allows systems and services to be accessible within an organization.
- The private cloud is operated only within a single organization. However, it may be managed internally or by third-party.
- The main advantage of these systems is that the enterprise retains full control over corporate data, security guidelines, and system performance.



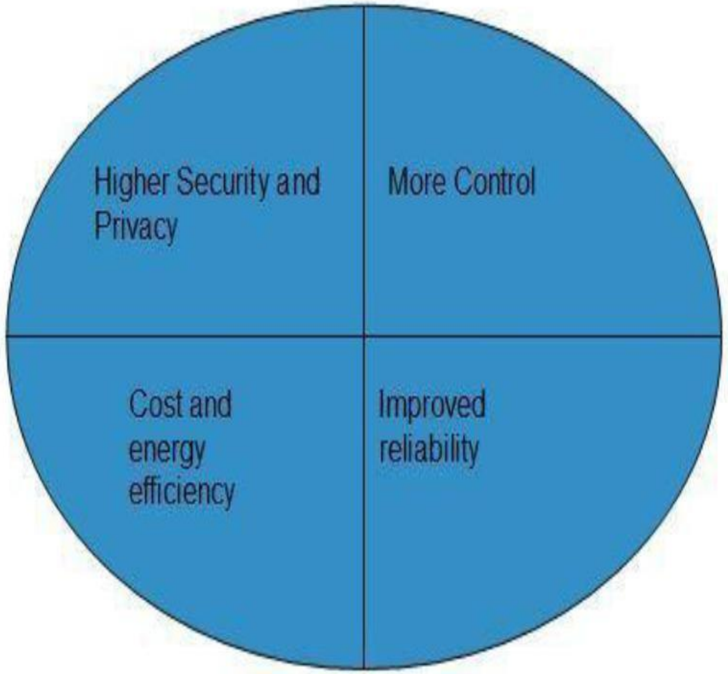
PRIVATE CLOUD

Advantages

- **Higher security and privacy -** Private cloud operations are not available to general public and resources are shared from distinct pool of resources, therefore, ensures high security and privacy.
- **More control -** Private clouds have more control on its resources and hardware because it is accessed only within an organization.
- **Cost and energy efficiency -** Private cloud resources are not as cost effective but they offer more efficiency than public cloud.
- **Improved Reliability**

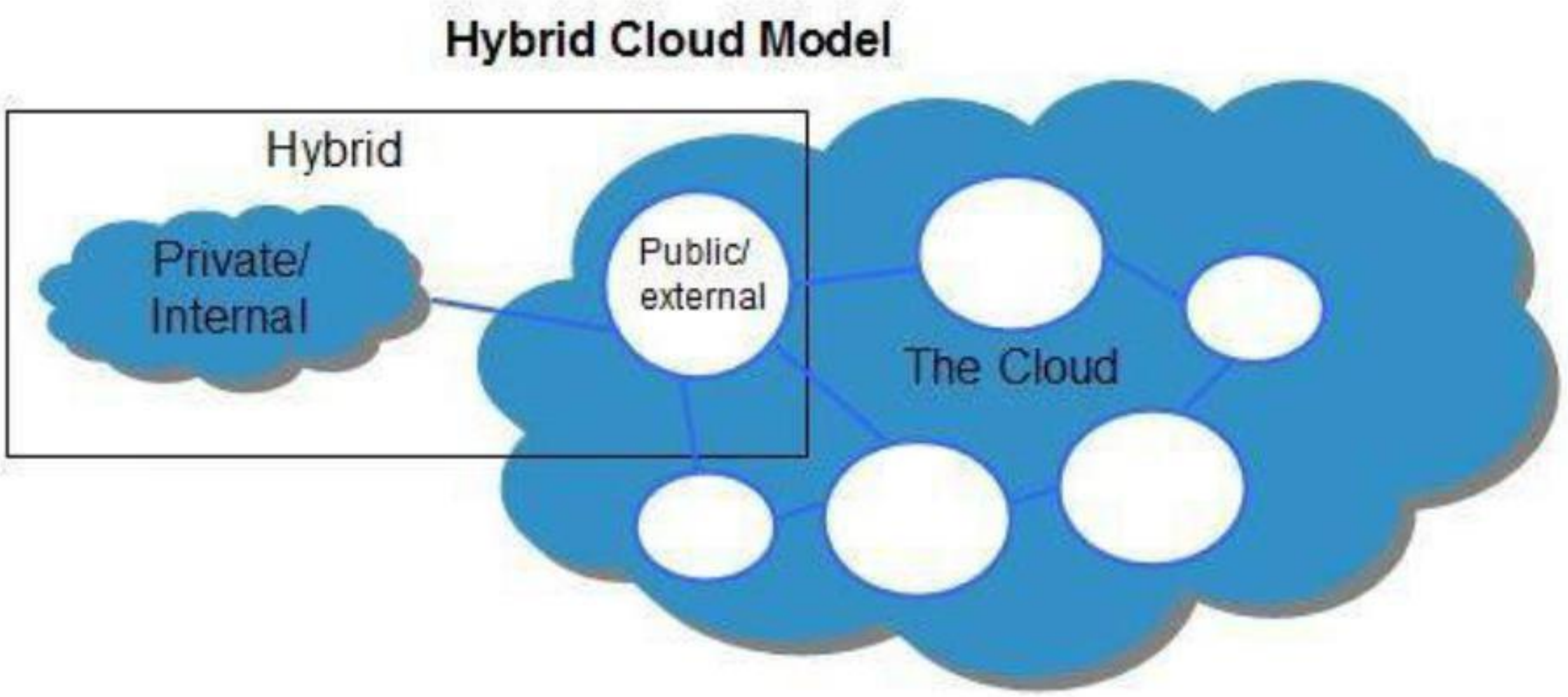
Disadvantages

- **Restricted area -** Only accessible locally and is difficult to deploy globally.
- **Inflexible pricing** – To meet demand, purchasing new hardware is very costly.
- **Limited scalability -** Can be scaled only within capacity of internal hosted resources.
- **Additional skills -** In order to maintain cloud deployment, organization requires more skilled and expertise.



HYBRID CLOUD

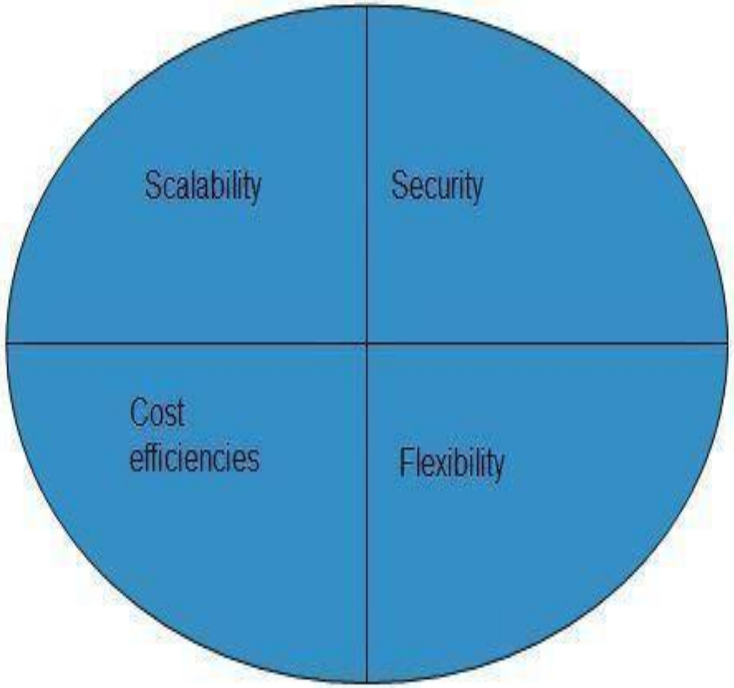
- This can be a combination of private and public clouds that support the requirement to **retain some data** in an organization, and also the **need** to **offer services** in the **cloud**.
- A company may use internal resources in a private cloud and maintain total control over its proprietary data. It can then use a public cloud storage provider for backing up less sensitive information.



HYBRID CLOUD

Advantages

- **Scalability -** It offers both features of public cloud scalability and private cloud scalability.
- **Flexibility -** It offers both secure resources and scalable public resources.
- **Cost efficiencies -** Public cloud are more cost effective than private, therefore hybrid cloud can have this saving.
- **Security -** Private cloud in hybrid cloud ensures higher degree of security.

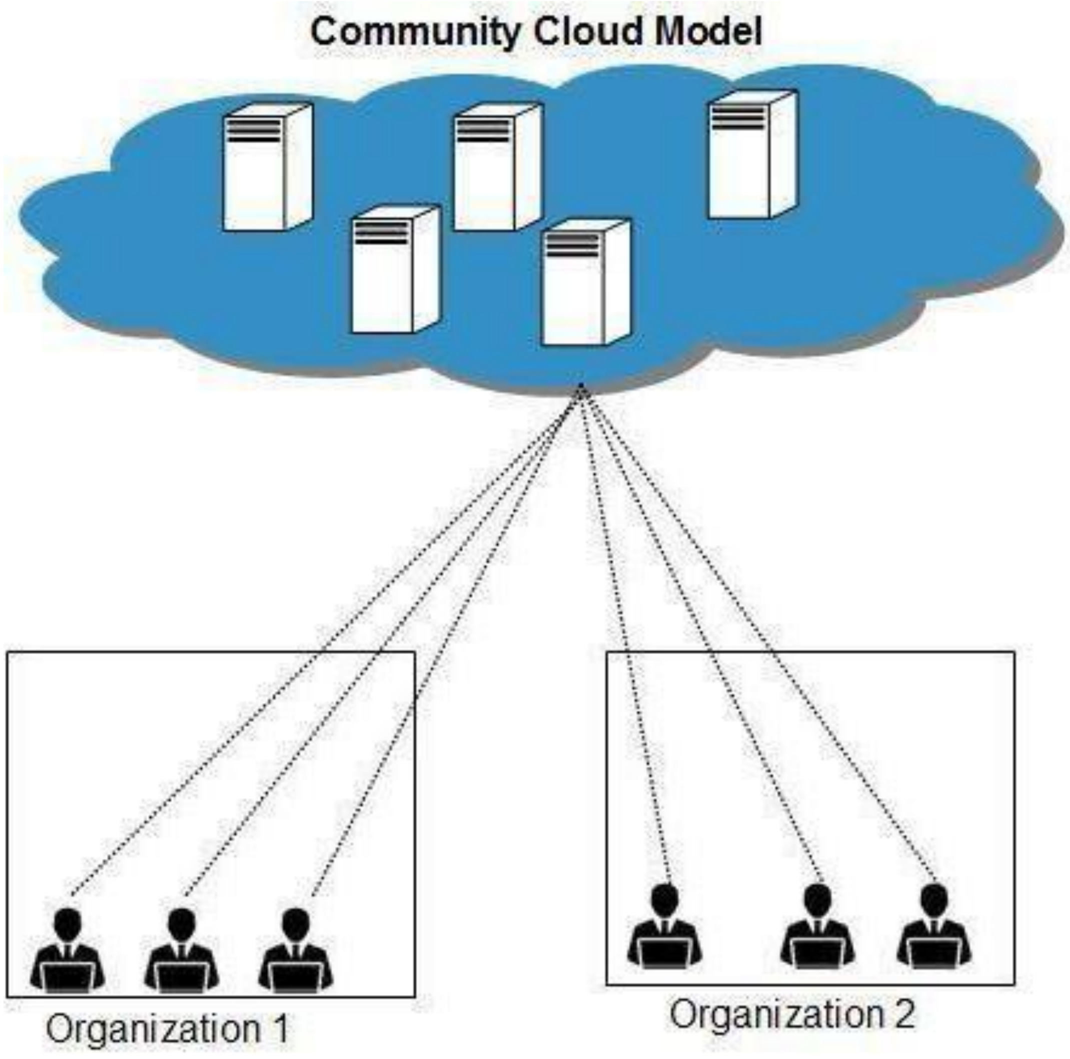


Disadvantages

- **Networking issues -** Networking becomes complex due to presence of private and public cloud.
- **Security compliance -** It is necessary to ensure that cloud services are compliant with organization's security policies.
- **Infrastructural dependency -** The hybrid cloud model is dependent on internal IT infrastructure, therefore it is necessary to ensure redundancy across data centers.

COMMUNITY CLOUD

- The community cloud allows system and services to be accessible by group of organizations.
- It shares the infrastructure between several organizations from a specific community.
- It may be managed internally or by the third-party.



COMMUNITY CLOUD

Advantages

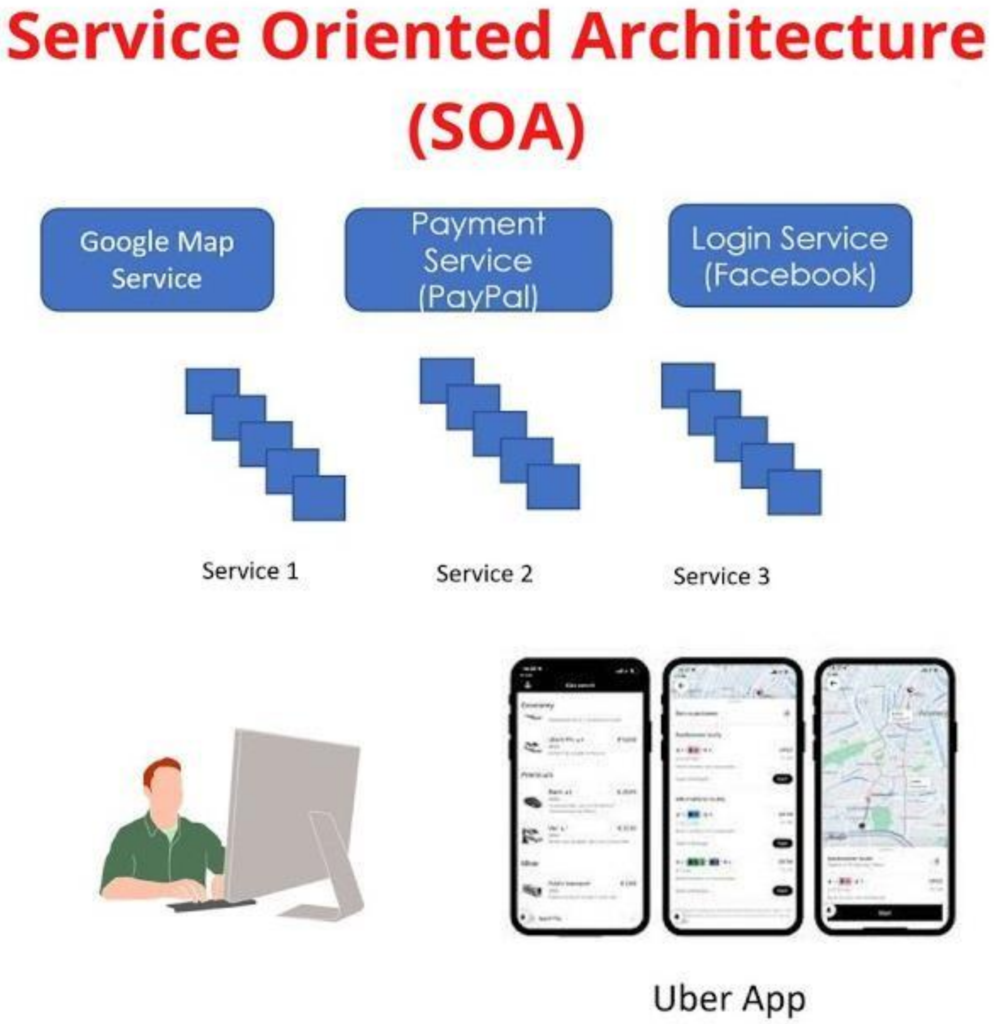
- **Cost effective –** It offers same advantage as that of public cloud at low cost. Sharing between organizations community cloud provides an infrastructure to share cloud resources and capabilities among several organizations.
- **Security -** It is comparatively more secure than the public cloud.

Disadvantages

- Since all data is housed at one location,it might be accessible by others.
- It is also challenging to allocate responsibilities of governance, security and cost.

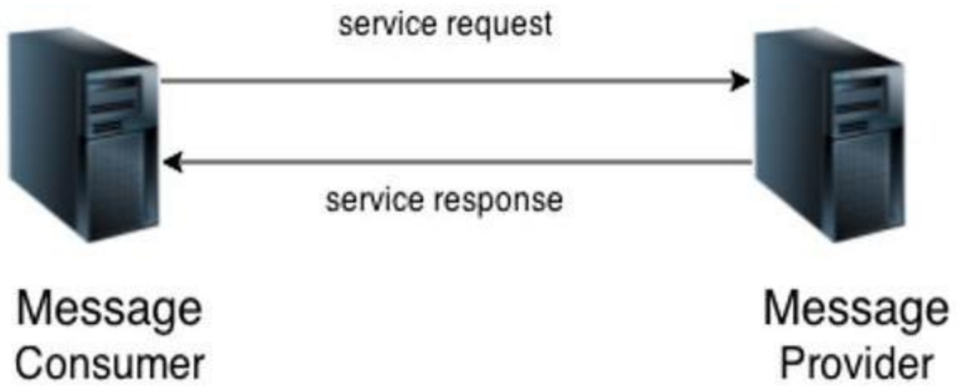
SERVICE ORIENTED ARCHITECTURE (SOA)

- Service-Oriented Architecture (SOA) is a software design approach that involves building software components as reusable services.
- Each service represents a specific business function and can be accessed over a network using standard protocols.
- In cloud design and implementation, SOA is used to create a flexible, scalable, and modular architecture that can support a wide range of applications and services.
- SOA enables the creation of loosely-coupled services that can be combined and orchestrated to support complex business processes.
- SOA is particularly well-suited for cloud computing because it allows for the creation of services that can be accessed from anywhere on the internet.
- This means that applications can be built using services from multiple sources, including public cloud providers, private cloud infrastructure, and legacy systems.



SERVICE ORIENTED ARCHITECTURE (SOA)

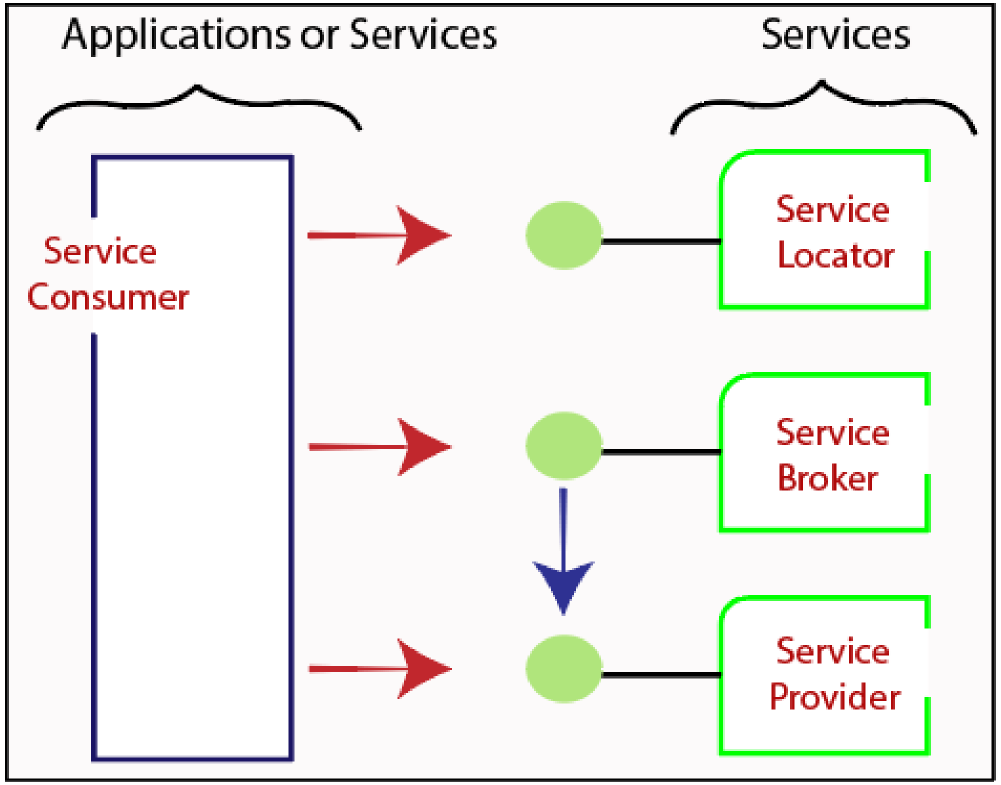
- **Service**
  - A service is a well-defined, self-contained function that represents a unit of functionality.
  - A service can exchange information from another service.
  - It is not dependent on the state of another service.
  - It uses a loosely coupled, message-based communication model to communicate with applications and other services.
- **Service Connections**
  - Service consumer sends a service request to the service provider, and the service provider sends the service response to the service consumer.
  - The service connection is understandable to both the service consumer and service provider.



SERVICE ORIENTED ARCHITECTURE (SOA)

Service-Oriented Terminologies

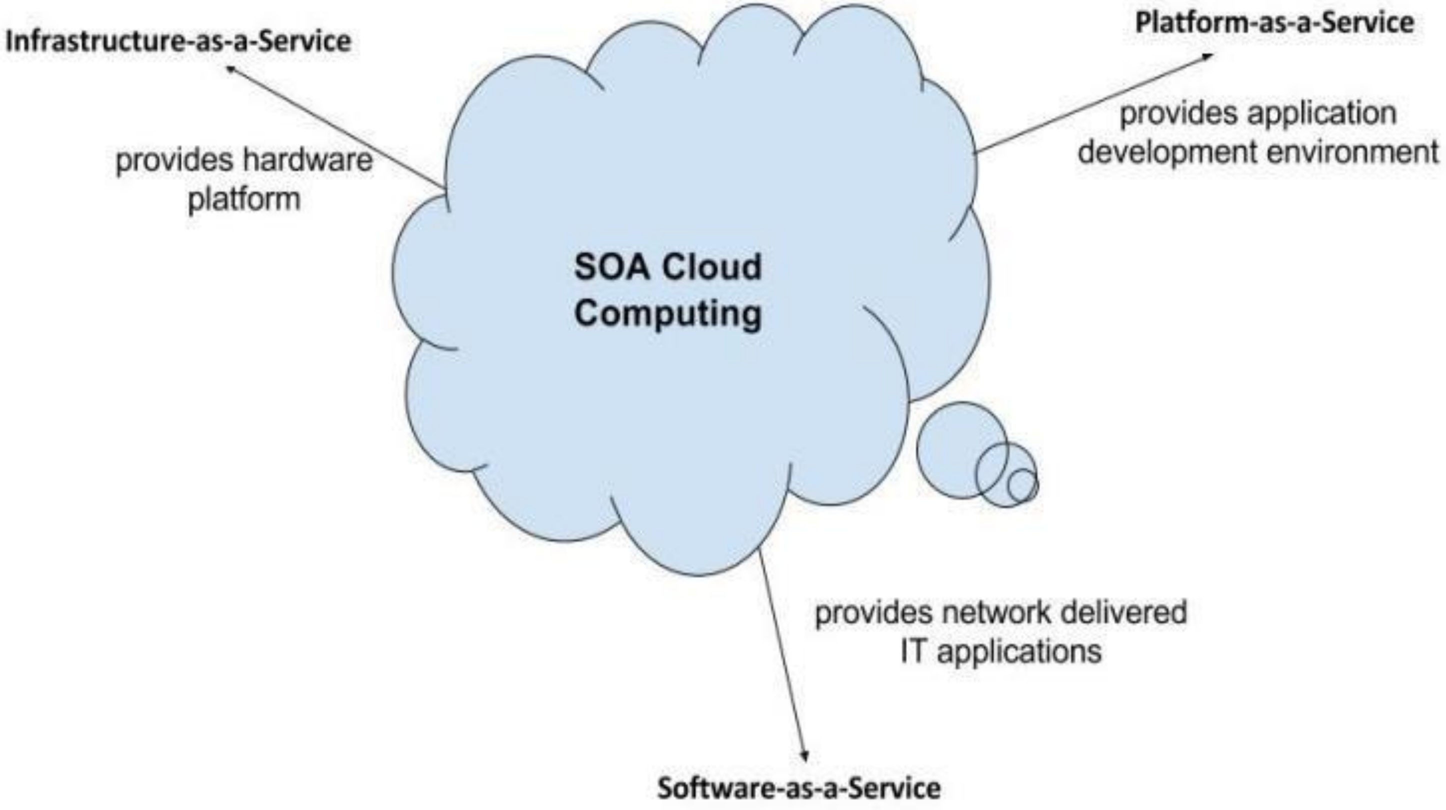
- **Services -** The **logical entities** defined by one or more published interfaces.
- **Service provider -** Software entity that **implements** a service specification.
- **Service consumer -** Requestor or client that **calls** a service provider. A service consumer can be another service or an end-user application.
- **Service locator -** A service provider that acts as a **registry** and is responsible for **examining service provider interfaces** and service locations.
- **Service broker -** A service provider that **pass service requests** to one or more additional service providers.



SERVICE ORIENTED ARCHITECTURE (SOA)

Characteristics of SOA

- Are loosely coupled.
- Support interoperability.
- Are location-transparent
- Are self-contained.

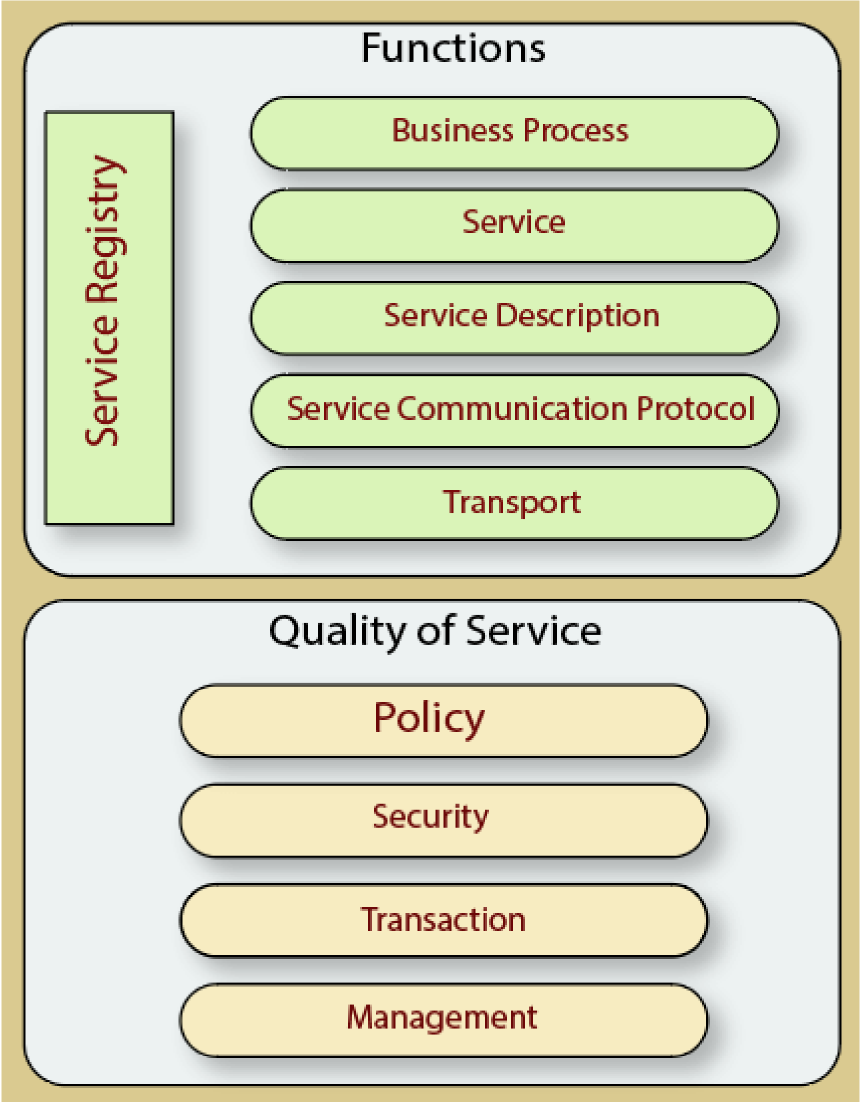


SOA cloud computing along with the models

SERVICE ORIENTED ARCHITECTURE (SOA)

Components of service-oriented architecture

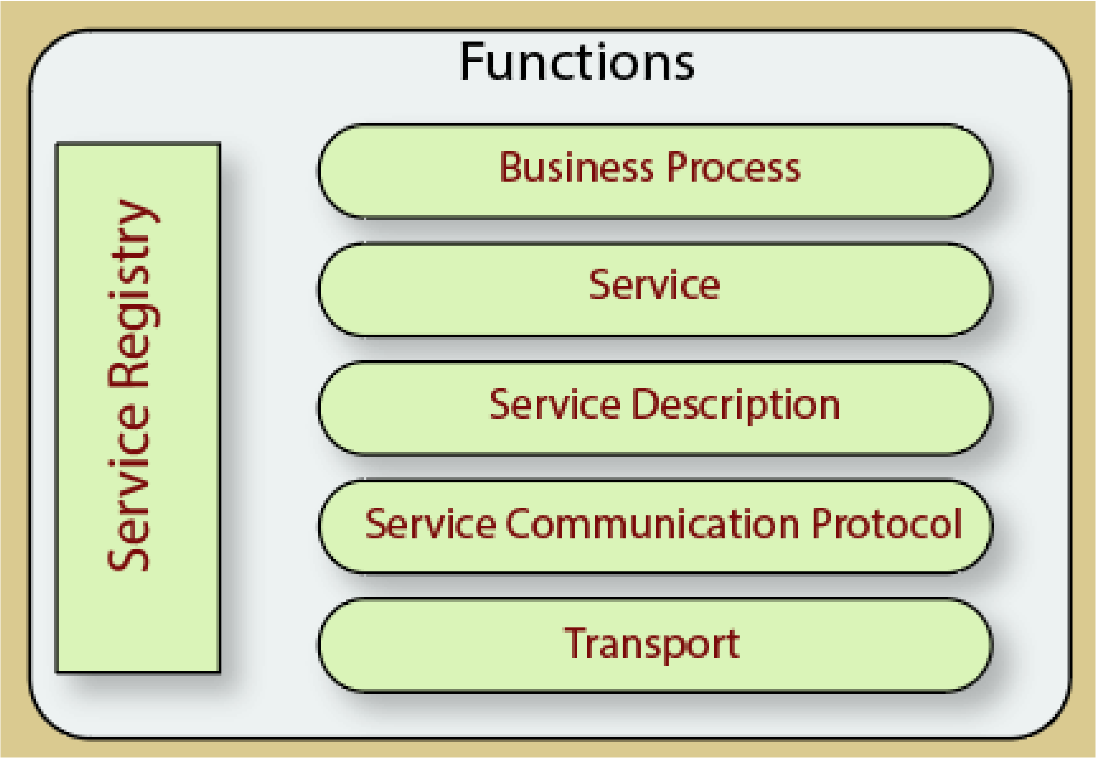
- The service-oriented architecture stack can be categorized into two parts –
    - **Functional Aspects**
    - **Quality of Service Aspects**



SERVICE ORIENTED ARCHITECTURE (SOA)

Functional Aspects

- **Transport -** It transports the **service requests** from the service consumer to the service provider and **service responses** from the service provider to the service consumer.
- **Service Communication Protocol -** It allows the service provider and the service consumer to **communicate** with each other.
- **Service Description -** It **describes** the **service** and **data** required to invoke it.
- **Service -** It is an actual service.
- **Business Process -** It **represents** the **group of services** called in a particular sequence associated with the particular rules to meet the business requirements.
- **Service Registry -** It contains the **description** of data which is used by service providers to publish their services.



SERVICE ORIENTED ARCHITECTURE (SOA)

Quality of Service Aspects

- **Policy -** It represents the **set of protocols** according to which a service provider **makes** and **provide** the services to consumers.
- **Security -** It represents the **set of protocols** required for **identification** and **authorization**.
- **Transaction -** It provides the **surety** of consistent result.
  - This means, if we use the group of services to complete a business function, either all must complete or none of the complete.
- **Management -** It **defines** the **set of attributes** used to manage the services.

SERVICE ORIENTED ARCHITECTURE (SOA)

Advantages of SOA

- **Easy to integrate -** The integration is a service specification that provides implementation transparency.
- **Manage Complexity -** Due to service specification, the complexities get isolated, and integration becomes more manageable.
- **Platform Independence -** The services are platform-independent as they can communicate with other applications through a common language.
- **Loose coupling -** It facilitates to implement services without impacting other applications or services.
- **Parallel Development -** As SOA follows layer-based architecture, it provides parallel development.
- **Available -** The SOA services are easily available to any requester.
- **Reliable -** As services are small in size, it is easier to test and debug them.

SERVICE ORIENTED ARCHITECTURE (SOA)

- Practical Applications of SOA:
- SOA is used in many ways around us whether it is mentioned or not.

1. SOA infrastructure is used by many armies and air force to deploy situational awareness systems.

2. SOA is used to improve the healthcare delivery.

3. Nowadays many apps are games and they use inbuilt functions to run.

   For example, an app might need GPS so it uses inbuilt GPS functions of the device. This is SOA in mobile solutions.

4. SOA helps maintain museums a virtualized storage pool for their information and content.

PRACTICAL SECURITY , TRUST AND PRIVACY

- In the modern digital world, effective sharing of information between individuals and organizations has become a critical requirement. This increases the demand for data sharing and privacy. The presence of personal information such as medical records, financial records and school records have been identified as a major barrier to data sharing. This limits the sharing of data for different purposes, such as academic or academic research which are important to support various activities in society such as improving public health care and policy making.
- Sharing of data effectively and without any revelation of personal information is still a major challenge.
- Several approaches, such as anonymization and encryption, have emerged to solve this problem, but this is achieved with a significant loss of information. There is therefore a problem of sharing the micro data while protecting the data. The main challenge when disclosing information is to provide as much information as possible while ensuring the confidentiality of an individual. This means that limiting disclosure of shared data requires careful consideration between data utility and individual privacy.
- This problem can be represented by asking the following main questions:
  - How can we ensure privacy in a cloud environment while reducing the loss of information? What approaches can be put in place to reduce the amount of information loss while striving to protect the privacy of the individual in a cloud environment?
  - How to design, develop and implement anonymization approaches to improve data privacy and utility in a distributed environment and especially in a cloud mode? What approaches can be put in place to reduce the amount of information loss while striving to protect the privacy of the individual in a cloud environment?
  - How to design, develop and implement anonymization approaches to improve data privacy and utility in a distributed environment and especially in a cloud mode?

PRACTICAL SECURITY ,TRUST AND PRIVACY

Ways to ensure the Privacy and Confidentiality of data.

- **Maintain Trust –**
  - In the cloud industry, privacy and security of personal data are still among the barriers to adoption. Even if most companies use cloud technologies and consumers are increasingly adopting cloud-based products and services, it is still very difficult to know exactly if the data stored online is secure. In highly sensitive sectors such as health and finance, this uncertainty represents a major obstacle to the adoption and implementation of Cloud solutions. Adherence to a set of standards may be insufficient in the face of the most advanced issues of confidentiality but will in fact stimulate longer-term adoption in the most vulnerable sectors would enjoy the benefits of the cloud.

- **Ensure Confidentiality –**
  - Confidentiality ensures that customer data is accessible only by authorized entities. Different cloud computing solutions include privacy mechanisms such as identity and access management, encryption, and anonymization. The most secure access controls have no protection against an attacker gaining access to information, identification or keys. Thus, credentials or key management information are essential links in the design of security.

THANK YOU !